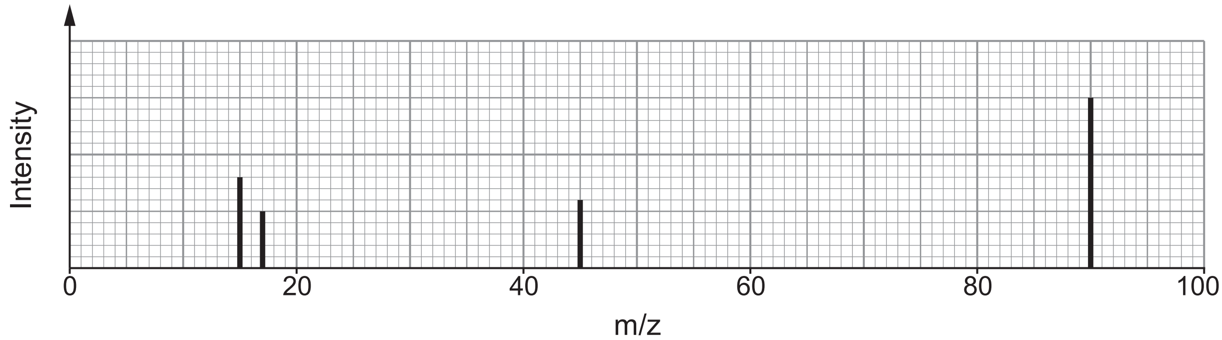
11. *(a)* Compound **X** contains only carbon, hydrogen and oxygen.

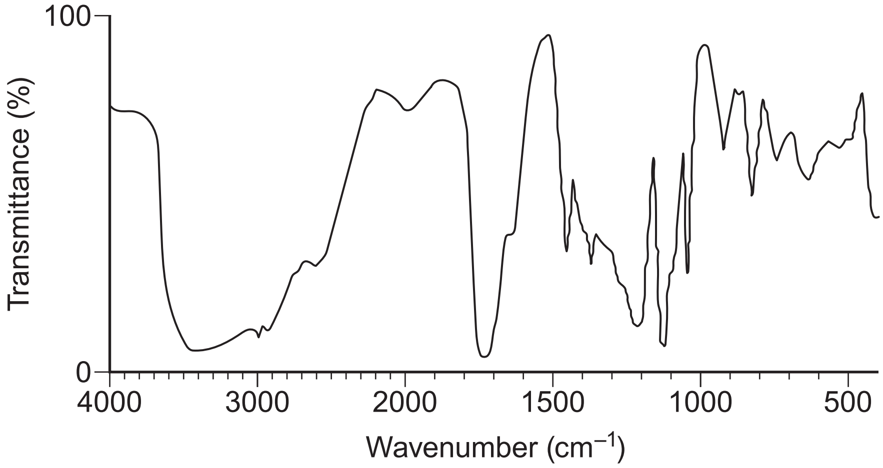
On analysis **X** was found to contain 40.0 % carbon and 6.67 % hydrogen by mass.

A simplified form of the mass spectrum, the IR spectrum and the low resolution ^{1}H NMR spectrum of **X** are shown.

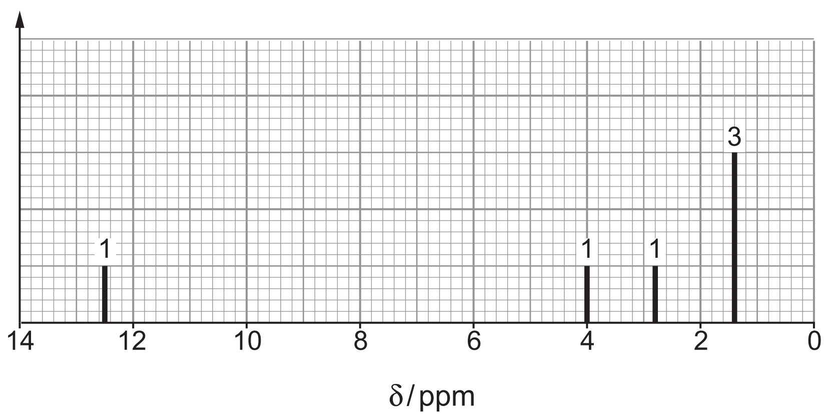
Mass spectrum



IR spectrum



^{1}H NMR spectrum



The numbers above the peaks show the relative areas of the peaks.

 (2410U20-1)

Examiner only

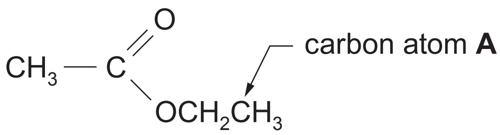
Use this information to identify compound **X**.

You must use information from **all** the sources given and explain how you used it. [10]

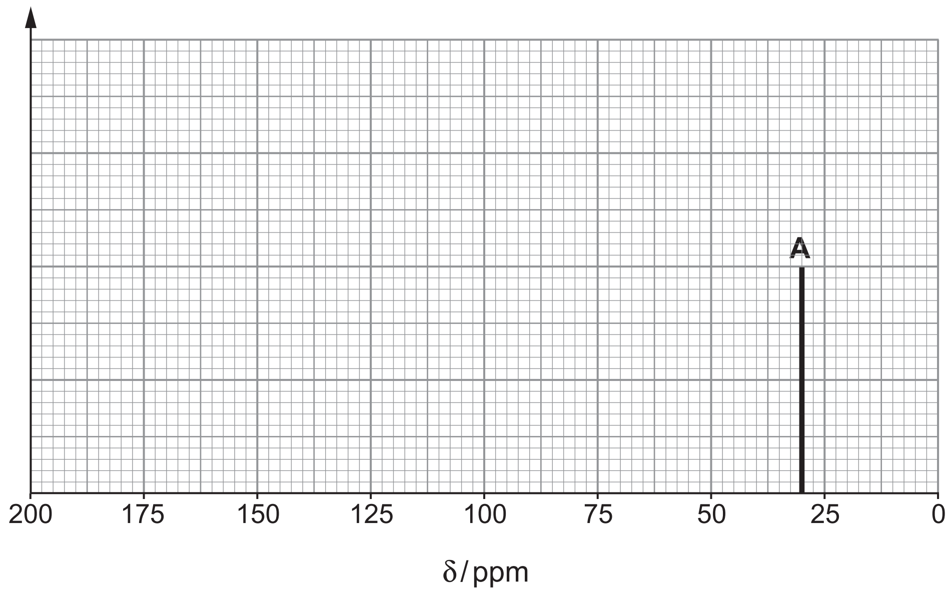
Examiner only

(b) Information about the structure of organic compounds can also be found using a ^{13}C NMR spectrum.

(i) On the axes below sketch the ^{13}C NMR spectrum that you would expect to obtain from ethyl ethanoate.



Label the diagram to show the species causing each peak.
The peak caused by carbon atom **A** is already included. [3]



(ii) A student said that valuable information about the nature of a sample being analysed could be obtained from the peak heights in both the ^{13}C and ^{1}H NMR spectra.

Comment on whether the student is correct. Give a reason for your answer. [1]

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END OF PAPER